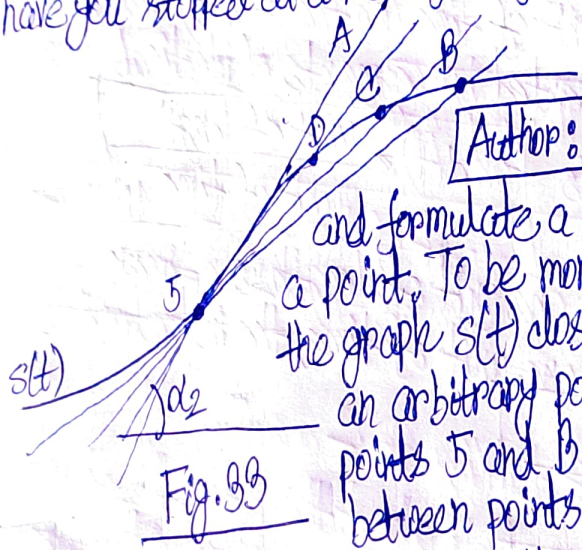


Reader: In Fig. 32 you have demonstrated a "straightening" of the graph by increasing its scale. But this straightening is only approximate. Why have you stopped at a mere four-fold scale increase?



Author: We can get rid of this approximation and formulate a more rigorous definition of a slope at a point. To be more specific, we consider a segment of the graph $s(t)$ close to point 5. In this segment we select an arbitrary point B and draw a secant through points 5 and B (Fig. 33). Next, on the same graph between points 5 and B we select an arbitrary point C and draw a new secant 5C. Further, we select an arbitrary point D in the segment between 5 and C and draw a new secant 5D. We may continue this process infinitely long and, as a result, we obtain a sequence of secants which converges to a certain straight line (line 5A in Fig. 33). This straight line is said to be tangent to the curve at point 5. The slope of the tangent is said to be the slope of the graph at a given point.

Reader: If I understand you correctly we are now in a position to formulate strictly the answer to the question about the instantaneous velocity.

Author: Try to do it, then.

Reader: The instantaneous velocity of a body at a moment of time t is the rate of change of $s(t)$ at the moment t . Numerically it is equal to the tangent of the slope of the tangent line to the graph of the function $s(t)$ at the moment t .

Author: Very good. But you should have mentioned that $s(t)$ expresses the distance covered by the body as a function of time.

Reader: This is true, my definition of the instantaneous velocity is tied to the graph of $s(t)$. What if the function $s(t)$ is not defined